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054323 - INTERNET OF THINGS

POLITECNICO DI MILANO

COMPUTER SCIENCE AND ENGINEERING

IoT Challenge 4 - Homework
Exercise 2

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1 Exercise 2

1.1 Problem Description

A nursing home uses a BLE-based IoT system to help caregivers locate the dental prostheses of Alzheimer patients. Each denture contains a small BLE beacon that transmits advertising packets periodically; BLE gateways distributed throughout the facility receive these packets and estimate the approximate position of the denture using RSSI measurements.
The current firmware follows a fixed-interval beaconing strategy:

```
loop forever:
    send_BLE_beacon()
    sleep(10 seconds)
```

This approach transmits at the same rate regardless of the actual state of the denture, leading to unnecessary energy expenditure. The objective is to design an improved firmware and sensing architecture that extends battery lifetime while fully preserving the localization service.

1.2 Key Observation

The localization service exists to help caregivers find a denture **when it is not being worn**. When the prosthesis is in the patient's mouth it cannot be lost, so frequent BLE advertising during that time is entirely wasteful. This motivates a context-aware design with three distinct operating states based on two physical conditions:
(i) whether the denture is in the mouth
(ii) whether it is moving

1.3 Proposed Solution

1.3.1 Three-State Logic

- **IN_MOUTH**: the denture is being worn. Localization is not needed. A very infrequent heartbeat beacon (every 600 s) is sent only to confirm that the device is still alive and reachable by the infrastructure.
- **STATIC**: the denture is outside the mouth and has been motionless for longer than a timeout T_{timeout}. It is likely resting on a known surface (nightstand, bathroom shelf). A slow beaconing rate (every 300 s) is sufficient to provide coarse localization on demand.

- **MOVING**: the denture is outside the mouth and in motion. It may be being searched for, handled, or displaced. Full beaconing rate (every 10 s) is used to support accurate RSSI-based localization.

The state machine is shown in Figure 1.

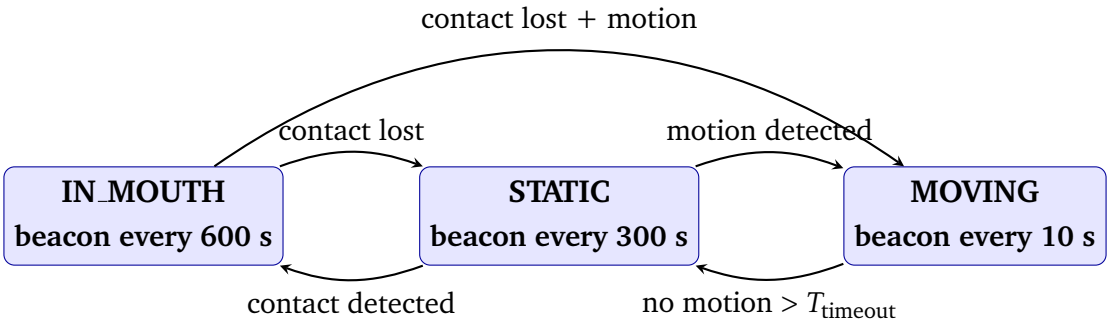


Figure 1: Three-state machine of the improved beaconing firmware

1.4 Additional Sensors

Two additional ultra-low-power sensors are integrated into the denture beacon module.

1.4.1 Sensor 1 - Capacitive Contact Sensor: TI FDC1004

The **Texas Instruments FDC1004** is a 4-channel capacitive sensor specifically designed for proximity and contact detection. When the denture is in the mouth, the surrounding tissue alters the measured capacitance above a configurable threshold, allowing reliable in-mouth / out-of-mouth detection.

Feature	Value
Supply voltage	3.3 V
Operating current	750 μA (active), 10 μA (standby)
Interface	I^2C
Measurement range	$\pm$15 pF offset + up to 100 pF
Package	WSON-8 (2$\times$2 mm)

Table 1: Key specifications of the TI FDC1004 capacitive sensor

The sensor is sampled at a low rate (e.g., once every 5 s) by the MCU to minimise its energy contribution. When not sampling, it is kept in standby mode (10 μA).

1.4.2 Sensor 2 - Accelerometer: ST LIS2DH12

The **ST LIS2DH12** is a 3-axis MEMS accelerometer that provides hardware motion-detection interrupts with extremely low idle power.

Feature	Value
Supply voltage	1.71–3.6 V
Current in low-power mode (1 Hz)	2 μA
Motion detection interrupt	Hardware (INT1 pin)
Configurable wake-up threshold	Yes (in mg steps)
Package	LGA-12 (2×2×1 mm)

Table 2: Key specifications of the ST LIS2DH12 accelerometer

The accelerometer runs continuously in low-power mode at 1 Hz ODR (2 μA). A hardware interrupt on INT1 wakes the MCU instantly when motion is detected, without any polling overhead.

1.5 Firmware Parameters

Parameter	Symbol	Value
Beacon interval - IN_MOUTH state	T_{mouth}	600 s
Beacon interval - STATIC state	T_{static}	300 s
Beacon interval - MOVING state	T_{moving}	10 s
Motion timeout (MOVING → STATIC)	T_{timeout}	60 s
Contact sensor sampling period	T_{cap}	5 s
Accelerometer ODR	–	1 Hz

Table 3: Firmware configuration parameters

1.6 Improved Firmware - Pseudocode

```
// --- States ---
STATE_IN_MOUTH = 0
STATE_STATIC   = 1
STATE_MOVING   = 2

// --- Global variables ---
state            = STATE_MOVING    // conservative start
last_motion_time = now()
```

```
// ------------------------------------------------
// ISR: fired by LIS2DH12 INT1 on motion detection
// ------------------------------------------------
function motion_interrupt_handler():
    last_motion_time = now()
    if state == STATE_STATIC:
        state = STATE_MOVING

// ------------------------------------------------
// Initialization
// ------------------------------------------------
function app_main():
    initialize_accelerometer(
        mode      = LOW_POWER,
        odr       = 1 Hz,
        threshold = 0.1 g,
        interrupt = motion_interrupt_handler
    )
    initialize_capacitive_sensor(
        mode    = STANDBY,
        channel = 0
    )

    // Main control loop
    loop forever:

        // --- 1. Read contact sensor to detect in-mouth state ---
        contact = read_capacitive_sensor()    // wakes FDC1004 briefly

        if contact == IN_MOUTH:
            state = STATE_IN_MOUTH
        else:
            // Out of mouth: check motion
            if state == STATE_IN_MOUTH:
                state = STATE_MOVING          // just removed: assume motion
            if state == STATE_MOVING:
                if (now() - last_motion_time) > T_timeout:
                    state = STATE_STATIC

        // --- 2. Transmit beacon ---
        send_BLE_beacon()
```

```
// --- 3. Sleep according to current state ---
if state == STATE_IN_MOUTH:
    deep_sleep(T_mouth)       // 600 s, wake on cap sensor poll
else if state == STATE_STATIC:
    deep_sleep(T_static)      // 300 s, wake on timer or INT1
else:  // STATE_MOVING
    deep_sleep(T_moving)      // 10 s,  wake on timer or INT1
```

1.6.1 Logic Description

1. At each wake-up the MCU first samples the capacitive sensor to determine whether the denture is in the mouth.
2. If contact is detected (`STATE_IN_MOUTH`), a single low-frequency heartbeat beacon is sent and the device sleeps for 600 s. No localization-quality update is needed.
3. If contact is lost and motion is detected (`STATE_MOVING`), the device beacons every 10 s, enabling accurate RSSI-based localization.
4. If contact is lost and no motion has been detected for $T_{\text{timeout}} = 60$ s (`STATE_STATIC`), the device reduces its beacon rate to one every 300 s.
5. A hardware interrupt from the accelerometer can transition the device from `STATE_STATIC` to `STATE_MOVING` at any time, without waiting for the next scheduled wake-up.

1.7 Commercial Components

Component	Role	Key Specs
Nordic nRF52810	BLE SoC (MCU + radio)	4.6 mA TX, deep sleep 1.5 μA
ST LIS2DH12	Motion detection	2 μA LP, HW interrupt
TI FDC1004	In-mouth contact detection	10 μA standby, I^2C
Panasonic CR2032	Primary cell battery	225 mAh, 3 V, coin cell

Table 4: Commercial components for the proposed system

1.8 Energy Saving Estimate

1.8.1 Time Assumptions: When Is Localization Required?

The localization service is only needed when the denture is **not worn**, i.e., outside the patient's mouth. Based on the typical daily routine of an Alzheimer patient in a nursing home, the following time split is assumed:

State	Duration/day	Localization needed?	Rationale
IN_MOUTH	16 h	No	Worn during all waking hours
STATIC	7 h	Yes (coarse)	Overnight storage, placed during meals
MOVING	1 h	Yes (precise)	Handled, cleaned, searched for
Total localization time	**8 h/day**	**Yes**	Out of mouth: STATIC + MOVING

Table 5: Daily time split and localization requirements per state

The denture therefore requires active localization coverage for **8 hours per day** on average. During the remaining 16 hours (in-mouth), only a rare heartbeat beacon is sent to confirm device reachability, but no position estimate is needed.

1.8.2 Other Assumptions

- **Energy per BLE beacon**: $E_{\text{beacon}} = 15\ \mu\text{J}$, derived from a typical BLE advertising current of 15 mA, supply voltage of 3 V and transmission duration of 1 ms:

$$E_{\text{beacon}} = I_{\text{TX}} \times V \times t_{\text{TX}} = 15\ \text{mA} \times 3\ \text{V} \times 1\ \text{ms} = 15\ \mu\text{J} \tag{1}$$

- **Deep-sleep baseline** (MCU + LIS2DH12 + FDC1004 standby): $\approx 13.5\ \mu$A. This quiescent current is present in both the original and the proposed solution and therefore cancels in the energy saving comparison.
- Ideal channel: no retransmissions modelled.
- Gateway infrastructure is mains-powered and not considered.

1.8.3 Original Fixed-Rate Solution

The original firmware (the one given in the problem statement) beacons every 10 s unconditionally, 24 hours/day, with no awareness of whether the denture is worn or at rest:

$$N_{\text{orig}} = \frac{24 \times 3600}{10} = 8640 \quad \text{beacons/day} \tag{2}$$

$$E_{\text{day}}^{\text{orig}} = 8640 \times 15\ \mu\text{J} = \mathbf{129.6}\ \text{mJ/day} \tag{3}$$

1.8.4 Improved Solution

IN_MOUTH state - 16 h/day, heartbeat beacon every 600 s:

$$N_{\text{mouth}} = \frac{16 \times 3600}{600} = 96 \quad \Rightarrow \quad E_{\text{mouth}} = 96 \times 15\ \mu\text{J} = 1.44\ \text{mJ} \tag{4}$$

STATIC state - 7 h/day, beacon every 300 s:

$$N_{\text{static}} = \frac{7 \times 3600}{300} = 84 \quad \Rightarrow \quad E_{\text{static}} = 84 \times 15\ \mu\text{J} = 1.26\ \text{mJ} \tag{5}$$

MOVING state - 1 h/day, beacon every 10 s:

$$N_{\text{moving}} = \frac{1 \times 3600}{10} = 360 \quad \Rightarrow \quad E_{\text{moving}} = 360 \times 15\ \mu\text{J} = 5.4\ \text{mJ} \tag{6}$$

Total daily energy (improved solution):

$$E_{\text{day}}^{\text{improved}} = E_{\text{mouth}} + E_{\text{static}} + E_{\text{moving}} = 1.44 + 1.26 + 5.4 = \mathbf{8.1}\ \text{mJ/day} \tag{7}$$

1.8.5 Energy Saving vs. Original Fixed-Rate Solution

$$\Delta E = E_{\text{day}}^{\text{orig}} - E_{\text{day}}^{\text{improved}} = 129.6 - 8.1 = 121.5\ \text{mJ/day} \tag{8}$$

$$\text{Saving} = \frac{\Delta E}{E_{\text{day}}^{\text{orig}}} \times 100 = \frac{121.5}{129.6} \times 100 \approx \mathbf{93.7}\% \tag{9}$$

1.8.6 Battery Lifetime

Using a Panasonic CR2032 ($C = 225$ mAh, $V = 3$ V):

$$E_{\text{battery}} = 225 \times 10^{-3} \times 3 \times 3600 = 2430\ \text{J} = 2430000\ \text{mJ} \tag{10}$$

$$\text{Lifetime}_{\text{orig}} = \frac{2430000}{129.6} \approx 18750\ \text{days} \approx \mathbf{51}\ \text{years} \tag{11}$$

$$\text{Lifetime}_{\text{improved}} = \frac{2430000}{8.1} \approx 300000\ \text{days} \approx \mathbf{820}\ \text{years} \tag{12}$$

Note: in practice the CR2032 self-discharge limits realistic lifetime to 5-10 years regardless of load. The figures above are load-limited bounds; the key result is the ∼94% reduction in radio-related energy consumption, which makes battery depletion due to beaconing essentially negligible compared to self-discharge.

Result - Energy Saving vs. Original Fixed-Rate Solution

Solution / State	Duration	Beacons/day	Energy
Original (fixed 10 s, always on)	24 h	8640	129.6 mJ
Improved - IN_MOUTH (600 s interval)	16 h	96	1.44 mJ
Improved - STATIC (300 s interval)	7 h	84	1.26 mJ
Improved - MOVING (10 s interval)	1 h	360	5.40 mJ
Improved total	24 h	**540**	**8.1 mJ**
Saving vs. original		**8100 beacons**	**121.5 mJ (93.7%)**

The context-aware firmware reduces daily beacon energy consumption by approximately **94%** compared to the original fixed-rate solution, by eliminating unnecessary transmissions while the denture is worn and preserving full localization capability during the 8 hours/day when it is actually needed.